

HIMANSHU BADHANI

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 [website](#)

RESEARCH INTERESTS

My primary interests are open quantum systems, with a focus on understanding thermodynamic and information-theoretic properties of noisy processes.

Topics that I am currently fascinated with:

- Non-Hermitian quantum mechanics, with a focus on the formalism of pseudo-Hermitian quantum mechanics.
- Quantum information theory, with a focus on information-theoretic properties of the quantum channels.
- Quantum thermodynamics of quantum channels and higher order quantum processes.
- Algebraic quantum mechanics and the structure of C^* -algebras.
- Geometric features in quantum mechanics, specifically formalizing thermodynamics in a “curved” Hilbert space.

Topics that I plan to explore in near future:

- Understanding the dynamical causal structure in many body systems.
- Quantum simulations using open quantum system techniques.
- Quantum biology, exploring whether NV centers can be made more sensitive for biological sensing using non-Hermitian techniques.

EMPLOYMENT

Visvesvaraya postdoctoral fellow at the Center for Quantum Science and Technology,
IIIT Hyderabad

Feb 2025–Present

EDUCATION

Ph.D.

2018-2025

The Institute of Mathematical Sciences, Chennai, India.

Thesis: *Subsystem dynamics in pseudo-Hermitian quantum mechanics*.

Submitted on November 27, 2024.

Defended on August 04, 2025.

Advisor: Dr. C M Chandrashekar.

BS-MS

2011 - 2016

Indian Institute of Science Education and Research (IISER), Pune, India.

Thesis: *Quantum fields in de-Sitter space*

CGPA: 8.2/10

Advisor: Dr. Alok Laddha, CMI Chennai

Gap year: In 2017-2018 I took a gap year and explored small scale farming in my village. I am passionate about reforestation efforts in the Himalayas and am experimenting with low intervention farming. I also got interested in the mathematical theory of finance and briefly joined the doctoral program in finance at IIM Kolkata, a premier management institute in India, in 2018.

PUBLICATIONS

[Google scholar](#) / [ArXiv](#)

1. **HB** and S. Das, Conditional channel entropy sets fundamental limits on thermodynamic quantum information processing, arXiv:2604.01217 [quant-ph] (2026)
2. **HB**, Dhanuja G. S, and S. Das, Thermodynamic work capacity of quantum information processing, arXiv:2510.23731 [quant-ph] (2025).
3. **HB**, Dhanuja G. S, and S. Das, Thermodynamics of quantum processes: An operational framework for free energy and reversible athermality, arXiv:2510.12790 [quant-ph] (2025).
4. **HB**, Dhanuja G. S., S. Choudhary, V. Anand, and S. Das. Erasure cost of a quantum process: a thermodynamic meaning of the dynamical min-entropy, *Quantum Sci. Technol.* **11** 015038, (2026).

5. **HB**, C. M. Chandrashekar. Reduced dynamics in quasi-Hermitian systems, *Annals of Physics*, Volume 484, 170304, (2026).
6. R. Basu, A. Chakraborty, **HB**, M. Alimuddin, and S. Bhattacharya. Exploring non-markovianity in ergodic channels: Measuring memory retention through ergotropy. *Phys. Rev. A* **111**, 032416, (2025).
7. F. Buscemi, R. Gangwar, K. Goswami, **HB**, T. Pandit, B. Mohan, S. Das, and M. N. Bera. Causal and noncausal revivals of information: A new regime of non-markovianity in quantum stochastic processes. *PRX Quantum* **6**, 020316, (2025).
8. **HB** and S. Ghosh. Decomposition of a system in pseudo-hermitian quantum mechanics. *J. Phys. A: Math. Theor.* **58** 165301, (2025).
9. **HB**, S. Banerjee, and C. M. Chandrashekar. Non-Hermitian quantum walks and non-Markovianity: the coin-position interaction. *Phys. Scr.* **99** 105112, (2024).
10. A. Mitra, **HB**, and S. Ghosh. Improvement in quantum communication using quantum switch. *Phys. Scr.* **98** 045101, (2023).
11. **HB** and C. M. Chandrashekar. Gravitationally induced entanglement dynamics between two quantum walkers. *Eur. Phys. J. C* **81**, 454, (2021).

CONTRIBUTED TALKS AND SEMINARS

1. (Upcoming) Physics seminar (online) at Heriot-Watt University, Edinburgh, June 2026. **Title:** Free energy of a quantum channel.
2. Physics seminar at IMSc Chennai, May 2026, **Title:** Thermodynamic aspects of quantum processes.
3. [Physics seminar](#) at TIFR Hyderabad, May 2026, **Title:** Thermodynamic aspects of quantum processes.
4. Physics seminar at IIIT Hyderabad, Jan. 2024, **Title:** Virtual subsystems of non-trivial inner product spaces.
5. [Chennai Strings Meeting 2023](#), IMSc, Chennai, **Title:** Partial tracing in a non-trivial metric Hilbert space.
6. [PHHQP20 2023](#), Koc University, Istanbul, **Title:** PT-symmetric quantum walk: exploring the reduced dynamics.
7. [PiQUST 2023](#) IIT Madras, **Title:** Non-Hermitian quantum walks.
8. [IMSc60 2023](#), IMSc Chennai, **Title:** Non-Hermitian quantum walks.

POSTER PRESENTATIONS IN INTERNATIONAL CONFERENCES

1. [TQC 2025](#) IISc Bangalore, **Title:** Thermodynamic interpretation of dynamical min-entropy.
2. [QCMC 2024](#), IIT Madras, **Title:** Defining subsystems in Hilbert spaces with non-Euclidean metric.
3. [PQIQC 2024](#), S. N. Bose NCBS, Kolkata, **Title:** Virtual sub-systems in a non-trivial metric Hilbert space.
4. [AQIS 2021](#), University of Tokyo, Japan (online), **Title:** Non-Hermitian quantum walks: position-coin interaction.
5. [Indo-German Workshop](#) on Quantum Science and Technologies 2020, IIT Madras, **Title:** Gravitationally induced entanglement between two quantum walkers.
6. [QFTA 2019](#), IISER Mohali, **Title:** Gravitationally induced entanglement between two quantum walkers.

RESEARCH VISITS

1. Research internship 2016–2017
Harish-Chandra Research Institute, Allahabad, India.
Project title: *de-Sitter stability and the inflationary potential*
Advisor: Dr. Dileep Jatkar
2. Visiting researcher, IIIT Hyderabad, Jan. 2024.
3. Visiting postdoc, IMSc Chennai, Nov. 2024-Jan. 2025, and May 2026.

TEACHING EXPERIENCE

- Currently mentoring two PhD students at IIIT Hyderabad
 - [1] Shradha Nair (2nd year): Witnessing quantum chaos through conditional entropy of channels.
 - [2] Dhanuja G S (4th year): Thermodynamics of higher order processes.
- Tutoring:
 - * Quantum Information Theory by Dr. Siddhartha Das, Monsoon 2025
 - * Introduction to Quantum Information by Dr. C. M. Chandrashekar, Spring 2020
- Mentored a few undergraduate summer students at IMSc over the years.

PEER REVIEW AND OUTREACH

- **Peer review:** *Class. Quantum Grav., J. Phys. A: Math. Theor., Physica Scripta, Quantum Inf Process, TQC* 2024.
- Panel member in the workshop on [Impact of Quantum Technologies on Life Sciences](#), organized with NITI Aayog, Gov. of India.
- YouTube manager for [Quantum Meets](#).
- Among the early members of [Disha](#) IISER Pune, a volunteer organization working on educational equity. I also started and ran a series of talks called *Talk for Twenty* and a magazine called *Pahal*, both of which are still going strong at IISER Pune.

LANGUAGES

Fluent: English, Hindi, Kumaoni
Beginner: Tamil, Marathi, German (A1 level)

ACADEMIC MERITS AND AWARDS

- [Visvesvaraya Post-Doctoral Fellowship](#) (2025-2026) by the Ministry of Electronics and Information Technology (MeitY), Gov. of India.
- Rank 22 in the **Joint Entrance Screening Test (JEST) 2018** for PhD positions in India.
- Qualified for the **Junior Research Fellowship** by CSIR, Gov. of India in 2016.
- *Excellent* grade for the MS thesis titled *Quantum fields in de-Sitter space*.
- INSPIRE fellowship (2011-2016) by the Dept. Of Science and Technology, Gov. of India

REFERENCES

Dr. C M Chandrashekar [✉](#)
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Bangalore
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Dr. Sibasish Ghosh [✉](#)
Institute of Mathematical Sciences,
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